My Spotify Wrapped 2025

by Elma Fortunate Phiri

Aim:

To analyse my Spotify 2025 streaming data in order to uncover personal listening patterns, favorite songs and artists and visualize trends over time

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import warnings
warnings.filterwarnings(action='ignore')

df= pd.read_json(r"c:\Users\My PC\Desktop\Datasets\Spotify Extended
Streaming History\Streaming_History_Audio_2024-2025_1.json")
df.head()
```

```
                      ts platform  ms_played conn_country
ip_addr  \
0  2024-08-13T18:09:27Z  windows     156500           IN
210.212.210.98
1  2024-08-13T18:12:18Z  windows     170986           IN
210.212.210.98
2  2024-08-13T18:14:39Z  windows     140247           IN
210.212.210.98
3  2024-08-13T18:17:31Z  windows     171902           IN
210.212.210.98
4  2024-08-13T18:19:32Z  windows     120080           IN
210.212.210.98

      master_metadata_track_name master_metadata_album_artist_name  \
0  Remember those quiet evenings                       Ivan Blois
1               The Last Goodbye                        Aquilegia
2                  Calolziocorte                   Grazia Bellini
3          Lost In The Bookstore                        Otto Noir
4                  A Day In June                       Half Giant

  master_metadata_album_album_name
spotify_track_uri  \
0    Remember those quiet evenings
spotify:track:6K6xmEal7cYw4UaWr5W88w
1                 The Last Goodbye
spotify:track:0CFHCZ9LKhFA1DqExQHkf9
2                    Calolziocorte
```

```
spotify:track:1eq8ybJ7diRj2rGsVwNVv6
3            Lost In The Bookstore
spotify:track:73uNzueWJEr3dwJ4OgcY3I
4                     A Day In June
spotify:track:0AlsRtqib2JbMHDWvcDrus

  episode_name  ... audiobook_uri audiobook_chapter_uri  \
0         None  ...           NaN                   NaN
1         None  ...           NaN                   NaN
2         None  ...           NaN                   NaN
3         None  ...           NaN                   NaN
4         None  ...           NaN                   NaN

   audiobook_chapter_title  reason_start  reason_end  shuffle skipped
offline  \
0                      NaN     trackdone   trackdone    False   False
False
1                      NaN     trackdone   trackdone    False   False
False
2                      NaN     trackdone   trackdone    False   False
True
3                      NaN     trackdone   trackdone    False   False
True
4                      NaN     trackdone   trackdone    False   False
True

   offline_timestamp  incognito_mode
0         1723572411           False
1         1723572568           False
2         1723572740           False
3         1723572880           False
4         1723573053           False

[5 rows x 23 columns]
```

```
df1= pd.read_json(r"c:\Users\My PC\Desktop\Datasets\Spotify Extended
Streaming History\Streaming_History_Audio_2025.json")
df1.head()
```

```
                      ts platform  ms_played conn_country
ip_addr  \
0  2025-09-02T16:59:41Z  android        760           ZW
102.128.79.206
1  2025-09-02T17:04:17Z  android     277051           ZW
102.128.79.206
2  2025-09-02T17:04:26Z  android       7470           ZW
102.128.79.206
3  2025-09-02T17:04:29Z  android       2027           ZW
102.128.79.206
4  2025-09-02T17:04:31Z  android       2028           ZW
```

```
102.128.79.206

  master_metadata_track_name master_metadata_album_artist_name  \
0              Ubani Obayazi                            Sbahle
1                     Umlilo                        Dr MaVibes
2                    Wake Up                           Teargas
3                      Imali                         Ami Faku
4                 Izishimane                      Blaq Diamond

  master_metadata_album_album_name
spotify_track_uri  \
0                    Ubani Obayazi
spotify:track:0zfiD5kOGI2nR7ediPAUAF
1                           Umlilo
spotify:track:2fo0KVxU9j24cpqziLnGsr
2                    Num8er Num8er
spotify:track:4cD1cQVU7CN2KDgcZ1azpk
3                            Imali
spotify:track:0mLgDscA9VfdA8K50BXQos
4                     Zulu Romance
spotify:track:36nCJkvY4WL3MzjJiafr19

   episode_name  ...  audiobook_uri  audiobook_chapter_uri  \
0           NaN  ...            NaN                    NaN
1           NaN  ...            NaN                    NaN
2           NaN  ...            NaN                    NaN
3           NaN  ...            NaN                    NaN
4           NaN  ...            NaN                    NaN

   audiobook_chapter_title  reason_start  reason_end  shuffle skipped
offline  \
0                      NaN        fwdbtn      fwdbtn    False    True
False
1                      NaN        fwdbtn   trackdone    False   False
False
2                      NaN     trackdone      fwdbtn    False    True
False
3                      NaN        fwdbtn      fwdbtn    False    True
False
4                      NaN        fwdbtn      fwdbtn    False    True
False

   offline_timestamp  incognito_mode
0         1756832380           False
1         1756832381           False
2         1756832657           False
3         1756832666           False
4         1756832669           False

[5 rows x 23 columns]
```

```
spotify= pd.concat([df,df1], ignore_index= True)
spotify.head()
```

```
                      ts platform  ms_played conn_country
ip_addr  \
0  2024-08-13T18:09:27Z  windows     156500           IN
210.212.210.98
1  2024-08-13T18:12:18Z  windows     170986           IN
210.212.210.98
2  2024-08-13T18:14:39Z  windows     140247           IN
210.212.210.98
3  2024-08-13T18:17:31Z  windows     171902           IN
210.212.210.98
4  2024-08-13T18:19:32Z  windows     120080           IN
210.212.210.98

      master_metadata_track_name master_metadata_album_artist_name  \
0  Remember those quiet evenings                       Ivan Blois
1               The Last Goodbye                        Aquilegia
2                  Calolziocorte                   Grazia Bellini
3          Lost In The Bookstore                        Otto Noir
4                  A Day In June                       Half Giant

  master_metadata_album_album_name
spotify_track_uri  \
0    Remember those quiet evenings
spotify:track:6K6xmEal7cYw4UaWr5W88w
1                  The Last Goodbye
spotify:track:0CFHCZ9LKhFA1DqExQHkf9
2                     Calolziocorte
spotify:track:1eq8ybJ7diRj2rGsVwNVv6
3            Lost In The Bookstore
spotify:track:73uNzueWJEr3dwJ4OgcY3I
4                     A Day In June
spotify:track:0AlsRtqib2JbMHDWvcDrus

  episode_name  ... audiobook_uri audiobook_chapter_uri  \
0         None  ...           NaN                   NaN
1         None  ...           NaN                   NaN
2         None  ...           NaN                   NaN
3         None  ...           NaN                   NaN
4         None  ...           NaN                   NaN

   audiobook_chapter_title  reason_start  reason_end  shuffle skipped
offline  \
0                      NaN     trackdone   trackdone    False   False
False
1                      NaN     trackdone   trackdone    False   False
False
2                      NaN     trackdone   trackdone    False   False
```

```
True
3                      NaN     trackdone   trackdone    False   False
True
4                      NaN     trackdone   trackdone    False   False
True

   offline_timestamp  incognito_mode
0         1723572411           False
1         1723572568           False
2         1723572740           False
3         1723572880           False
4         1723573053           False

[5 rows x 23 columns]
```

```
spotify.columns
```

```
Index(['ts', 'platform', 'ms_played', 'conn_country', 'ip_addr',
       'master_metadata_track_name',
'master_metadata_album_artist_name',
       'master_metadata_album_album_name', 'spotify_track_uri',
'episode_name',
       'episode_show_name', 'spotify_episode_uri', 'audiobook_title',
       'audiobook_uri', 'audiobook_chapter_uri',
'audiobook_chapter_title',
       'reason_start', 'reason_end', 'shuffle', 'skipped', 'offline',
       'offline_timestamp', 'incognito_mode'],
      dtype='object')
```

```
#Covert ts to datetime
spotify["ts"]= pd.to_datetime(spotify["ts"])
```

```
spotify_25= pd.DataFrame(spotify)
spotify_25
```

```
                              ts platform  ms_played conn_country  \
0     2024-08-13 18:09:27+00:00  windows     156500           IN
1     2024-08-13 18:12:18+00:00  windows     170986           IN
2     2024-08-13 18:14:39+00:00  windows     140247           IN
3     2024-08-13 18:17:31+00:00  windows     171902           IN
4     2024-08-13 18:19:32+00:00  windows     120080           IN
...                         ...      ...        ...          ...
18174 2025-11-18 15:24:36+00:00  android     113575           ZW
18175 2025-11-18 15:26:55+00:00  android     137484           ZW
18176 2025-11-18 15:26:57+00:00  android       2096           ZW
18177 2025-11-18 15:31:04+00:00  android     248144           ZW
18178 2025-11-18 15:57:07+00:00  android     195455           ZW

               ip_addr     master_metadata_track_name  \
0      210.212.210.98  Remember those quiet evenings
1      210.212.210.98                The Last Goodbye
```

```
2      210.212.210.98                  Calolziocorte
3      210.212.210.98          Lost In The Bookstore
4      210.212.210.98                  A Day In June
...               ...                            ...
18174  102.128.76.208         Overloading (OVERDOSE)
18175  102.128.76.208                       Kwalula
18176  102.128.76.208                  Dangerous II
18177  102.128.76.208                BROWN SKIN GIRL
18178  102.128.76.208                    Ngimtholile

      master_metadata_album_artist_name  \
0                            Ivan Blois
1                             Aquilegia
2                        Grazia Bellini
3                             Otto Noir
4                            Half Giant
...                                 ...
18174                            Mavins
18175                   Sun-El Musician
18176                         Usimamane
18177                          Blue Ivy
18178                     Maverick Muji

               master_metadata_album_album_name  \
0                Remember those quiet evenings
1                             The Last Goodbye
2                                Calolziocorte
3                        Lost In The Bookstore
4                                A Day In June
...                                        ...
18174                                Chapter X
18175                    To the World & Beyond
18176                             20th: Deluxe
18177  The Lion King: The Gift [Deluxe Edition]
18178                              Ngimtholile

                          spotify_track_uri episode_name  ...
audiobook_uri  \
0      spotify:track:6K6xmEal7cYw4UaWr5W88w         None  ...
NaN
1      spotify:track:0CFHCZ9LKhFA1DqExQHkf9         None  ...
NaN
2      spotify:track:1eq8ybJ7diRj2rGsVwNVv6         None  ...
NaN
3      spotify:track:73uNzueWJEr3dwJ4OgcY3I         None  ...
NaN
4      spotify:track:0AlsRtqib2JbMHDWvcDrus         None  ...
NaN
...                                     ...          ...  ...
...
```

```
18174  spotify:track:3ES1fBQACydIODlZmQS359          NaN  ...
NaN
18175  spotify:track:3Rq1cgwfOjGCH8oxb9jiOz          NaN  ...
NaN
18176  spotify:track:7A15WOGrEJLCAwOilxJyxN          NaN  ...
NaN
18177  spotify:track:0sCyaAxs2y2h85hEv6qXMO          NaN  ...
NaN
18178  spotify:track:4XQNZPkMyo1LxR3NYbVegt          NaN  ...
NaN

      audiobook_chapter_uri  audiobook_chapter_title  reason_start  \
0                       NaN                      NaN     trackdone
1                       NaN                      NaN     trackdone
2                       NaN                      NaN     trackdone
3                       NaN                      NaN     trackdone
4                       NaN                      NaN     trackdone
...                     ...                      ...           ...
18174                   NaN                      NaN       appload
18175                   NaN                      NaN     trackdone
18176                   NaN                      NaN        fwdbtn
18177                   NaN                      NaN        fwdbtn
18178                   NaN                      NaN     trackdone

       reason_end  shuffle skipped offline  offline_timestamp
incognito_mode
0       trackdone    False   False   False         1723572411
False
1       trackdone    False   False   False         1723572568
False
2       trackdone    False   False    True         1723572740
False
3       trackdone    False   False    True         1723572880
False
4       trackdone    False   False    True         1723573053
False
...           ...      ...     ...     ...                ...
...
18174   trackdone     True   False   False         1763478622
False
18175      fwdbtn     True    True    True         1763479476
False
18176      fwdbtn     True    True    True         1763479615
False
18177   trackdone     True   False    True         1763479617
False
18178      logout     True   False    True         1763479864
False

[18179 rows x 23 columns]
```

```
 #filter my data for Jan 2025 to Nov 2025
start= '2025-01-01'
end= '2025-11-30'

spotify_25= spotify[(spotify['ts']>= start)  & (spotify['ts']<= end)]
spotify_25= pd.DataFrame(spotify_25)
spotify_25.head()
```

```
                            ts platform  ms_played conn_country  \
7412 2025-01-01 05:53:50+00:00  android     237499           IN
7413 2025-01-01 05:57:25+00:00  android     214041           IN
7414 2025-01-01 06:15:31+00:00  android     113420           IN
7415 2025-01-01 10:23:32+00:00  android     341349           IN
7416 2025-01-01 10:31:03+00:00  android     450415           IN

                                      ip_addr  \
7412                            103.140.19.66
7413                            103.140.19.66
7414  2401:4900:6743:24b9:65d0:4c74:2ffe:da0b
7415  2401:4900:6743:24b9:65d0:4c74:2ffe:da0b
7416  2401:4900:6743:24b9:65d0:4c74:2ffe:da0b

                            master_metadata_track_name  \
7412                                          Esandleni
7413                                         Kuyabongwa
7414                                        Ungukuphila
7415            Shayi'Moto (feat. Seemah & Yanda Woods)
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...

     master_metadata_album_artist_name  \
7412                  Nontokozo Mkhize
7413                      ChubbyCheeks
7414                    Xolly Mncwango
7415                   Mellow & Sleazy
7416                         Mr Pilato

                       master_metadata_album_album_name  \
7412                                          Esandleni
7413                                         Kuyabongwa
7414                                   Jesus Is Enough
7415            Shayi'Moto (feat. Seemah & Yanda Woods)
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...

                         spotify_track_uri episode_name  ...
audiobook_uri  \
7412  spotify:track:7LTDINonhUaOiWTyN3AxCI         None  ...
NaN
7413  spotify:track:5126E37MXDlVwL3GNsFE81         None  ...
NaN
7414  spotify:track:53wwaMp15eGyE12Cm7sxLa         None  ...
```

```
NaN
7415  spotify:track:0AQfRLzd7pQZcLtfh9Onb8         None  ...
NaN
7416  spotify:track:44Z9yOoNfCafT8nPZCimMs         None  ...
NaN

     audiobook_chapter_uri  audiobook_chapter_title  reason_start
reason_end  \
7412                    NaN                      NaN       appload
endplay
7413                    NaN                      NaN     trackdone
trackdone
7414                    NaN                      NaN     trackdone
logout
7415                    NaN                      NaN      clickrow
endplay
7416                    NaN                      NaN       playbtn
endplay

      shuffle skipped offline  offline_timestamp  incognito_mode
7412    False    True   False         1735710591           False
7413    False   False   False         1735710831           False
7414    False   False   False         1735711045           False
7415    False    True   False         1735726636           False
7416    False    True   False         1735727012           False

[5 rows x 23 columns]
```

```
spotify_25.tail()
```

```
                              ts platform  ms_played conn_country  \
18174 2025-11-18 15:24:36+00:00  android     113575           ZW
18175 2025-11-18 15:26:55+00:00  android     137484           ZW
18176 2025-11-18 15:26:57+00:00  android       2096           ZW
18177 2025-11-18 15:31:04+00:00  android     248144           ZW
18178 2025-11-18 15:57:07+00:00  android     195455           ZW

              ip_addr master_metadata_track_name  \
18174  102.128.76.208     Overloading (OVERDOSE)
18175  102.128.76.208                   Kwalula
18176  102.128.76.208               Dangerous II
18177  102.128.76.208            BROWN SKIN GIRL
18178  102.128.76.208                Ngimtholile

      master_metadata_album_artist_name  \
18174                            Mavins
18175                   Sun-El Musician
18176                         Usimamane
18177                          Blue Ivy
18178                     Maverick Muji
```

```
               master_metadata_album_album_name  \
18174                                  Chapter X
18175                      To the World & Beyond
18176                               20th: Deluxe
18177  The Lion King: The Gift [Deluxe Edition]
18178                                Ngimtholile

                          spotify_track_uri episode_name  ...
audiobook_uri  \
18174  spotify:track:3ES1fBQACydIODlZmQS359          NaN  ...
NaN
18175  spotify:track:3Rq1cgwfOjGCH8oxb9jiOz          NaN  ...
NaN
18176  spotify:track:7A15WOGrEJLCAwOilxJyxN          NaN  ...
NaN
18177  spotify:track:0sCyaAxs2y2h85hEv6qXMO          NaN  ...
NaN
18178  spotify:track:4XQNZPkMyo1LxR3NYbVegt          NaN  ...
NaN

      audiobook_chapter_uri  audiobook_chapter_title  reason_start  \
18174                    NaN                      NaN       appload
18175                    NaN                      NaN     trackdone
18176                    NaN                      NaN        fwdbtn
18177                    NaN                      NaN        fwdbtn
18178                    NaN                      NaN     trackdone

       reason_end  shuffle skipped offline  offline_timestamp
incognito_mode
18174   trackdone     True   False   False         1763478622
False
18175      fwdbtn     True    True    True         1763479476
False
18176      fwdbtn     True    True    True         1763479615
False
18177   trackdone     True   False    True         1763479617
False
18178      logout     True   False    True         1763479864
False

[5 rows x 23 columns]
```

```
spotify_25.shape
```

```
(10767, 23)
```

```
spotify_25.columns
```

```
Index(['ts', 'platform', 'ms_played', 'conn_country', 'ip_addr',
       'master_metadata_track_name',
```

```
'master_metadata_album_artist_name',
       'master_metadata_album_album_name', 'spotify_track_uri',
'episode_name',
       'episode_show_name', 'spotify_episode_uri', 'audiobook_title',
       'audiobook_uri', 'audiobook_chapter_uri',
'audiobook_chapter_title',
       'reason_start', 'reason_end', 'shuffle', 'skipped', 'offline',
       'offline_timestamp', 'incognito_mode'],
      dtype='object')
```

```
#Renaming columns
spotify_25.rename(columns = {
                  'ts': 'Date',
                  'conn_country': 'Country',
                  'master_metadata_track_name': 'Track_name',
                  'master_metadata_album_artist_name':
'Album_artist_name',

'master_metadata_album_album_name':'Album_album_name'},
                  inplace= True
)

#Removing unwanted columns
spotify_25.drop(columns={'ip_addr','spotify_track_uri',
'episode_name',
       'episode_show_name', 'spotify_episode_uri', 'audiobook_title',
       'audiobook_uri', 'audiobook_chapter_uri',
'audiobook_chapter_title',
       'reason_start', 'reason_end', 'shuffle','offline_timestamp',
'incognito_mode'}, inplace= True)

spotify_25.head()
```

```
                          Date platform  ms_played Country  \
7412 2025-01-01 05:53:50+00:00  android     237499      IN
7413 2025-01-01 05:57:25+00:00  android     214041      IN
7414 2025-01-01 06:15:31+00:00  android     113420      IN
7415 2025-01-01 10:23:32+00:00  android     341349      IN
7416 2025-01-01 10:31:03+00:00  android     450415      IN

                                                Track_name
Album_artist_name  \
7412                                          Esandleni  Nontokozo
Mkhize
7413                                          Kuyabongwa
ChubbyCheeks
7414                                         Ungukuphila    Xolly
Mncwango
7415            Shayi'Moto (feat. Seemah & Yanda Woods)   Mellow &
Sleazy
```

```
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...         Mr
Pilato

                                          Album_album_name  skipped
offline
7412                                            Esandleni     True
False
7413                                           Kuyabongwa    False
False
7414                                     Jesus Is Enough    False
False
7415            Shayi'Moto (feat. Seemah & Yanda Woods)     True
False
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...     True
False
```

```
spotify_25.columns
```

```
Index(['Date', 'platform', 'ms_played', 'Country', 'Track_name',
       'Album_artist_name', 'Album_album_name', 'skipped', 'offline'],
      dtype='object')
```

```
#Covert milliseconds to minutes
spotify_25['ms_played']= spotify_25['ms_played']/6000
spotify_25.head()
```

```
                          Date platform  ms_played Country  \
7412 2025-01-01 05:53:50+00:00  android  39.583167      IN
7413 2025-01-01 05:57:25+00:00  android  35.673500      IN
7414 2025-01-01 06:15:31+00:00  android  18.903333      IN
7415 2025-01-01 10:23:32+00:00  android  56.891500      IN
7416 2025-01-01 10:31:03+00:00  android  75.069167      IN

                                                Track_name
Album_artist_name  \
7412                                            Esandleni  Nontokozo
Mkhize
7413                                           Kuyabongwa
ChubbyCheeks
7414                                          Ungukuphila    Xolly
Mncwango
7415            Shayi'Moto (feat. Seemah & Yanda Woods)   Mellow &
Sleazy
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...         Mr
Pilato

                                          Album_album_name  skipped
offline
7412                                            Esandleni     True
False
```

```
7413                                          Kuyabongwa    False
False
7414                                     Jesus Is Enough    False
False
7415            Shayi'Moto (feat. Seemah & Yanda Woods)     True
False
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...     True
False
```

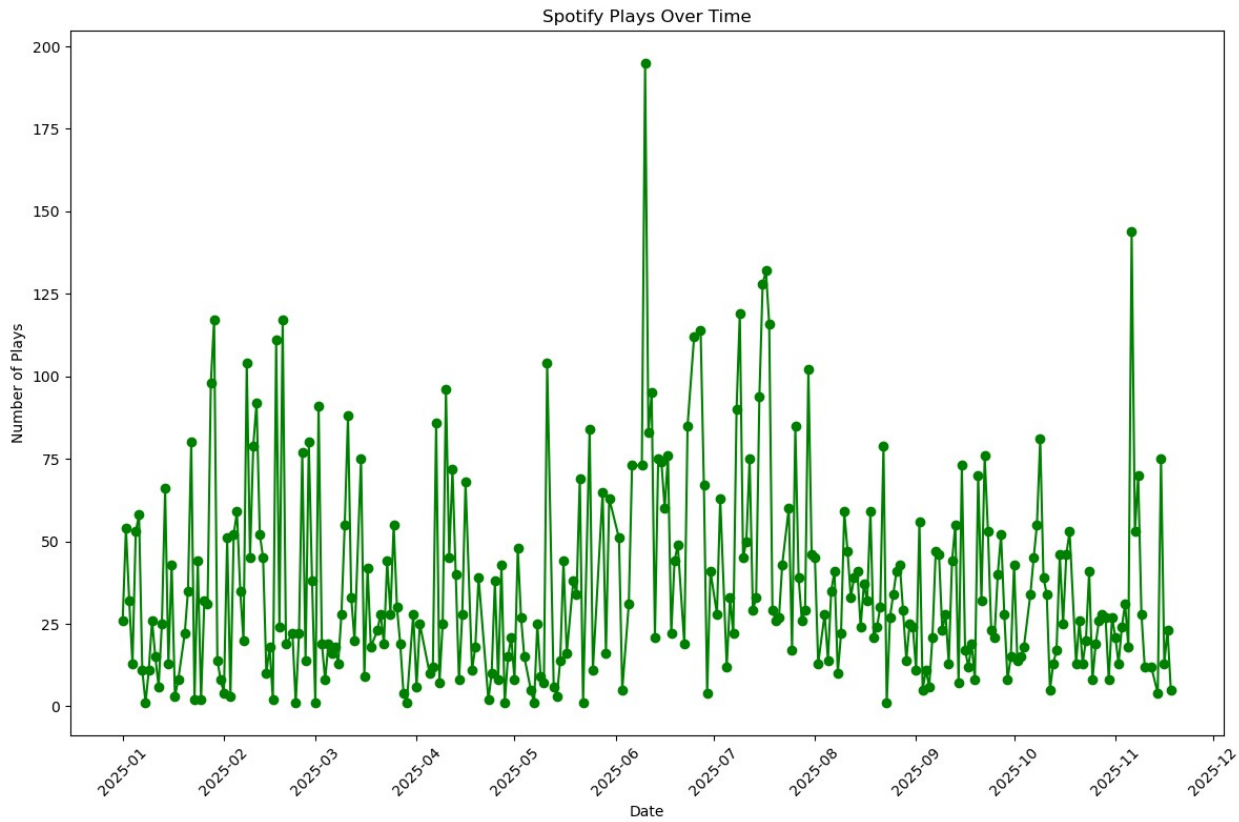
```
#Extract year, month, day, hour
spotify_25['Year']= spotify_25['Date'].dt.year
spotify_25['Month']= spotify_25['Date'].dt.month
spotify_25['Month_name']= spotify_25['Date'].dt.month_name()
spotify_25['Day']= spotify_25['Date'].dt.day
spotify_25['Hour']= spotify_25['Date'].dt.hour
```

```
spotify_25.head()
```

```
                          Date platform  ms_played Country  \
7412 2025-01-01 05:53:50+00:00  android  39.583167      IN
7413 2025-01-01 05:57:25+00:00  android  35.673500      IN
7414 2025-01-01 06:15:31+00:00  android  18.903333      IN
7415 2025-01-01 10:23:32+00:00  android  56.891500      IN
7416 2025-01-01 10:31:03+00:00  android  75.069167      IN

                                          Track_name
Album_artist_name  \
7412                                       Esandleni  Nontokozo
Mkhize
7413                                      Kuyabongwa
ChubbyCheeks
7414                                     Ungukuphila    Xolly
Mncwango
7415            Shayi'Moto (feat. Seemah & Yanda Woods)   Mellow &
Sleazy
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...         Mr
Pilato

                                     Album_album_name  skipped
offline  \
7412                                       Esandleni     True
False
7413                                      Kuyabongwa    False
False
7414                                     Jesus Is Enough    False
False
7415            Shayi'Moto (feat. Seemah & Yanda Woods)     True
False
```

```
7416  Biri Marung (Edit) (feat. Sje Konka, Focalisti...     True
False

      Year  Month Month_name  Day  Hour
7412  2025      1    January    1     5
7413  2025      1    January    1     5
7414  2025      1    January    1     6
7415  2025      1    January    1    10
7416  2025      1    January    1    10
```

```
#Number of tracks played per day

# Group by date and count plays
plays_per_day =
spotify_25.groupby(spotify_25['Date'].dt.date).size().reset_index(name
='Plays')

# Plot
plt.figure(figsize=(12,8))
plt.plot(plays_per_day['Date'], plays_per_day['Plays'], marker='o',
color= 'green')
plt.title("Spotify Plays Over Time")
plt.xlabel("Date")
plt.ylabel("Number of Plays")
plt.xticks(rotation=45)  # rotate dates for readability
plt.tight_layout()
plt.show()
```



My Listening Habits

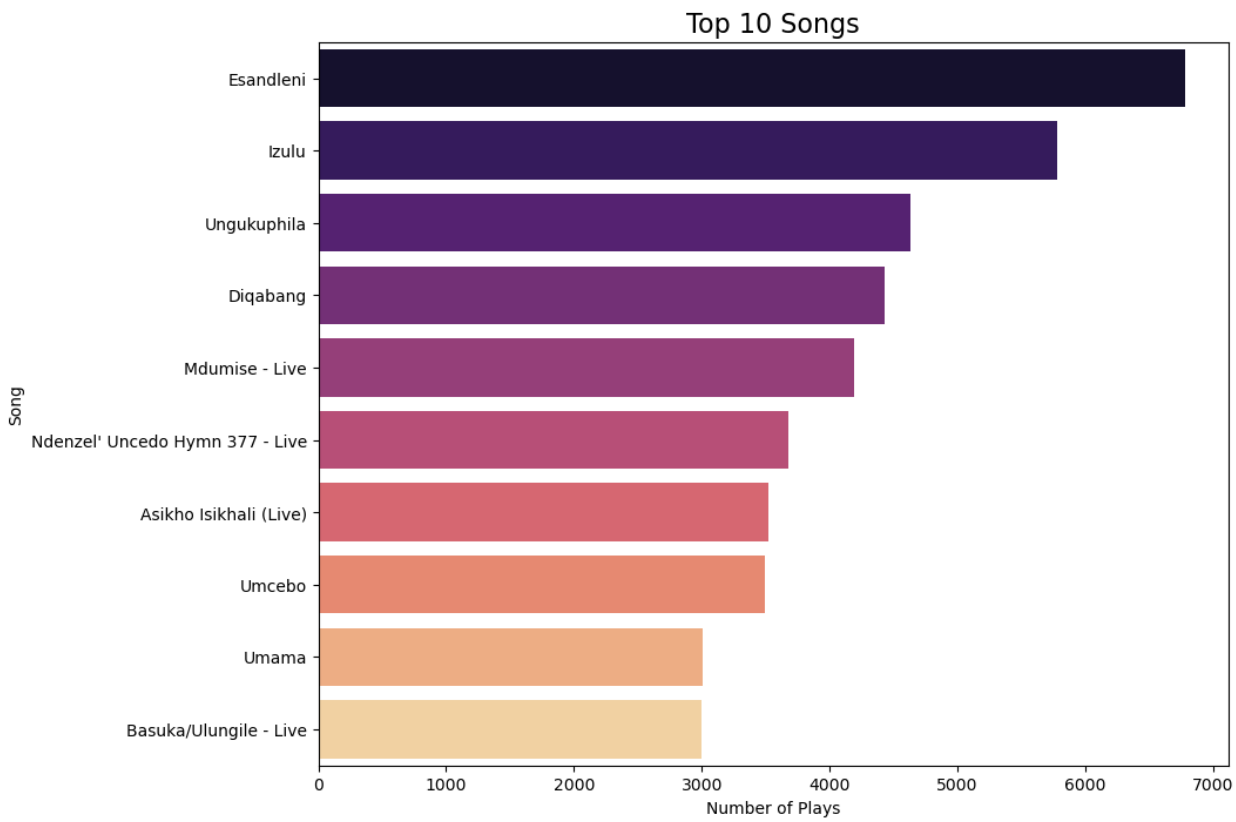
Top 10 Songs

```
#Most Songs Played
top_songs=  spotify_25.groupby('Track_name')
['ms_played'].sum().sort_values(ascending=False).head(10)
top_songs_df= pd.DataFrame(top_songs)
top_songs_df= top_songs.reset_index()
top_songs_df
```

```
                        Track_name    ms_played
0                        Esandleni  6778.049833
1                            Izulu  5776.951667
2                      Ungukuphila  4636.038000
3                         Diqabang  4426.301167
4                   Mdumise - Live  4194.513167
5  Ndenzel' Uncedo Hymn 377 - Live  3679.398000
6           Asikho Isikhali (Live)  3517.562500
7                           Umcebo  3498.146833
8                            Umama  3007.015333
9           Basuka/Ulungile - Live  2996.612167
```

```
plt.figure(figsize=(10,8))
sns.barplot(x=top_songs.values, y=top_songs.index, palette='magma')
```

```
plt.title("Top 10 Songs", fontsize= 16)
plt.xlabel("Number of Plays")
plt.ylabel("Song");
```

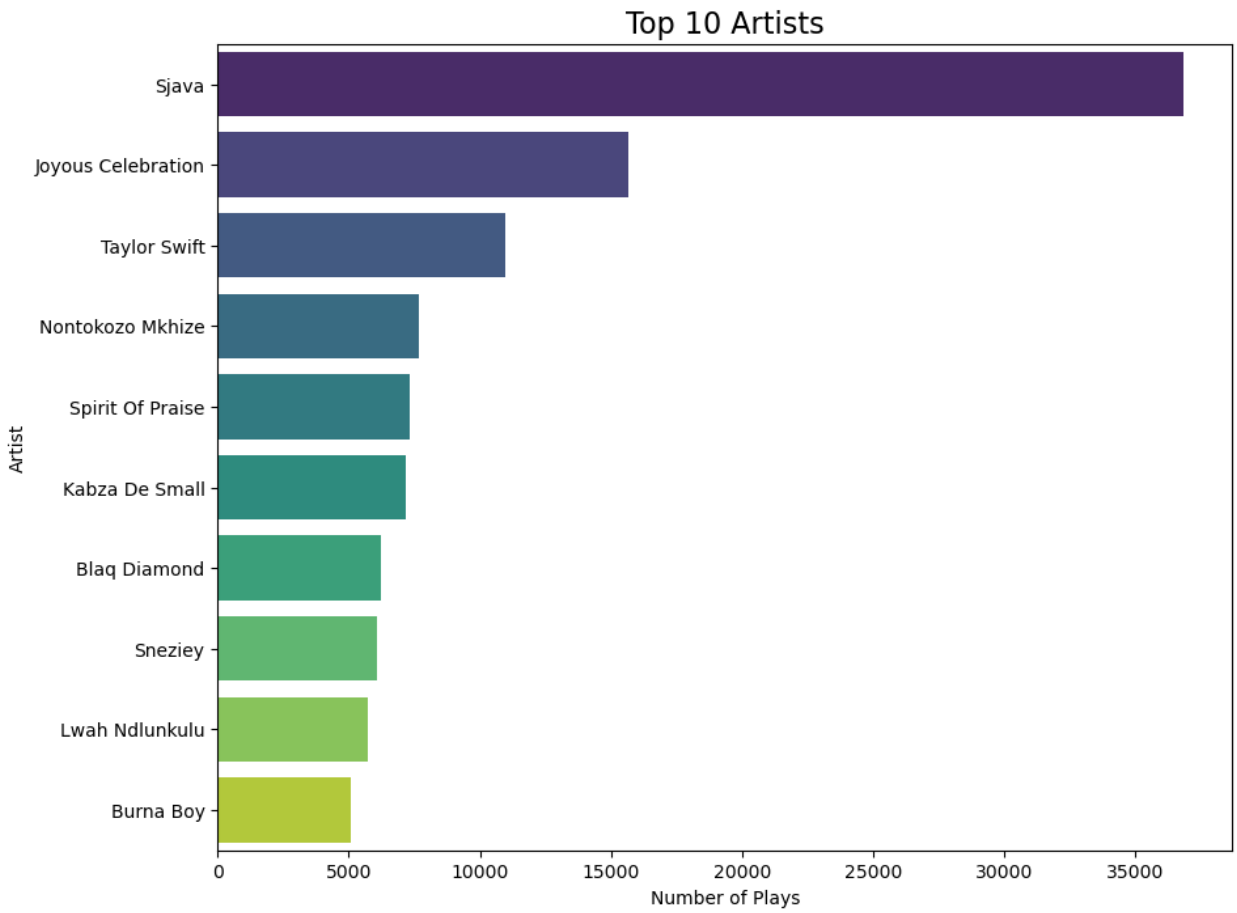


Top Artists

```
top_artist=  spotify_25.groupby('Album_artist_name')
['ms_played'].sum().sort_values(ascending=False).head(10)
top_artist_df= pd.DataFrame(top_artist)
top_artist_df
```

```
                        ms_played
Album_artist_name
Sjava                36857.167167
Joyous Celebration   15661.675333
Taylor Swift         10953.625167
Nontokozo Mkhize      7662.363333
Spirit Of Praise      7316.810167
Kabza De Small        7155.908333
Blaq Diamond          6225.223333
Sneziey               6062.186667
Lwah Ndlunkulu        5742.156667
Burna Boy             5085.743333
```

```
plt.figure(figsize=(10,8))
sns.barplot(x=top_artist.values, y=top_artist.index,
palette='viridis')
plt.title("Top 10 Artists", fontsize= 16)
plt.xlabel("Number of Plays")
plt.ylabel("Artist");
```



Top Albums

```
top_album=  spotify_25.groupby('Album_album_name')
['ms_played'].sum().sort_values(ascending=False).head(10)
top_album_df= pd.DataFrame(top_album)
top_album_df
```

```
                      ms_played
Album_album_name
Umqhele            7342.440167
Inkanyezi          6545.358667
Izulu              5891.311333
Inkanyezi 2.0      5503.453000
Imizwa             5468.514833
```

```
Esandleni          4710.814167
Jesus Is Enough    4659.654000
Umsebenzi          4587.169833
Amagama            4579.047333
Diqabang           4466.220500
```

Top 10 Songs by my favourite Artist

```
fav_artist= 'Sjava'
top_songs_artist=
spotify_25[spotify_25['Album_artist_name']==fav_artist].groupby('Track
_name')['ms_played'].sum().sort_values(ascending=
False).head(10).reset_index()
top_songs_artist
```

```
                                          Track_name    ms_played
0                                   Mdumise - Live  4194.513167
1                                           Umcebo  3498.146833
2                                            Umama  2989.784167
3                                     Uyena - Live  2460.326167
4                                      Kuye - Live  1893.994167
5                                            Linda  1876.944167
6                                  Sleepover - Live  1801.464000
7  Typhoon (feat. Sastii, LaCabra, Lowfeye, Blue ...  1593.287667
8                                       Confession  1542.903833
9                                         Valelise  1097.579500
```

How many unique songs did I listen to

```
unique_songs= spotify_25["Track_name"].nunique()

print("Count of Unique SOngs listened to in 2025: ", unique_songs)
```

```
Count of Unique SOngs listened to in 2025:  1978
```

```
#Unique_ Artists
unique_artists= spotify_25["Album_artist_name"].nunique()
print("Count of Unique Artist listened to in 2025: ", unique_artists)
```

```
Count of Unique Artist listened to in 2025:  718
```

```
#Song Diversity Ratio
total_plays= len(spotify_25)
song_diversity_ratio= unique_songs/total_plays
print("Song Diversity Ratio:",round(song_diversity_ratio *100, 2))
```

```
SOng Diversity Ratio: 18.37
```

```
#Longest Listening Session
longest_track= spotify_25.loc[spotify_25['ms_played'].idxmax()]
```

```
print(longest_track[['Track_name', 'Album_artist_name', 'ms_played',
'Date']])
```

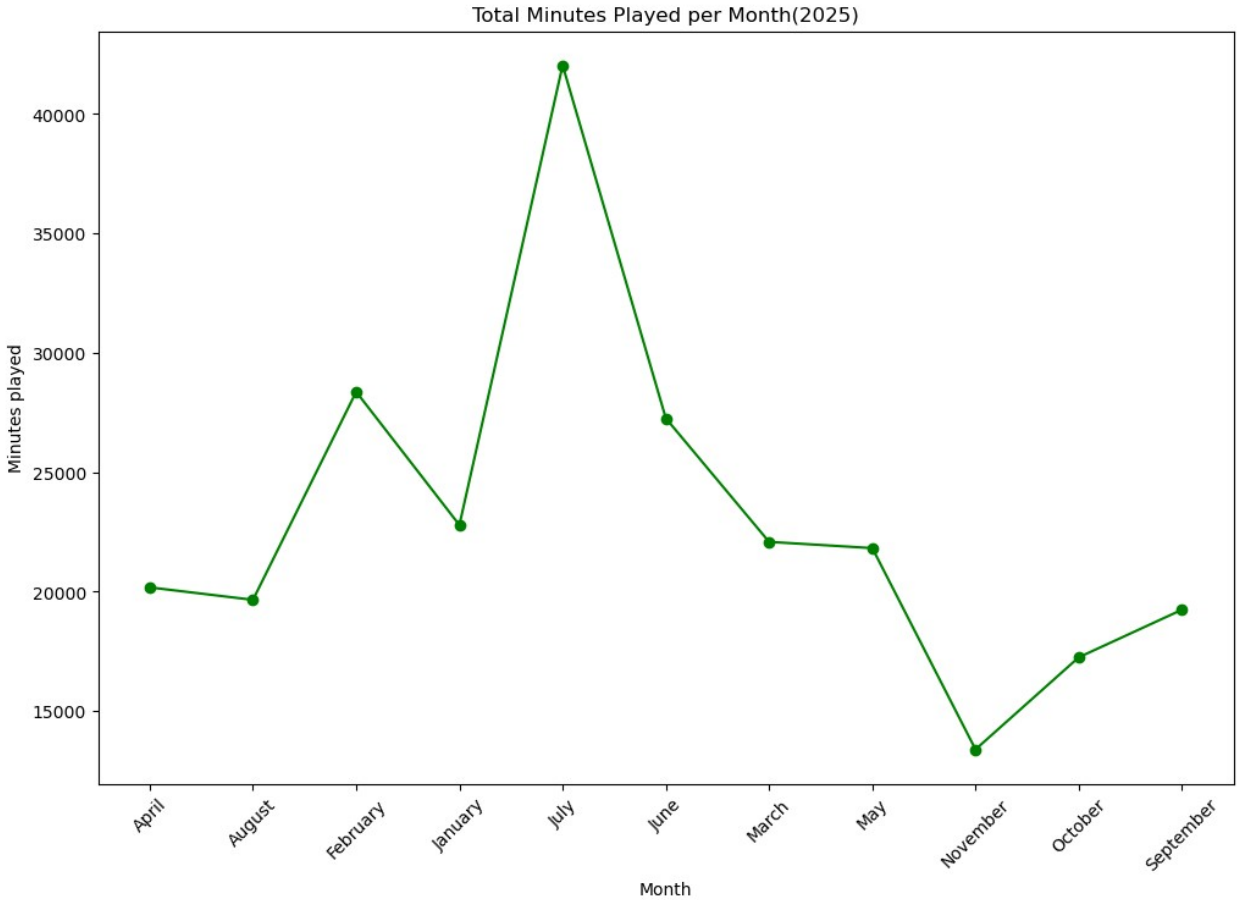
```
Track_name                        Mdumise - Live
Album_artist_name                          Sjava
ms_played                               293.2255
Date                  2025-10-13 16:43:47+00:00
Name: 17189, dtype: object
```

Listening Over Time

```
monthly_play= spotify_25.groupby('Month_name')
['ms_played'].sum().sort_values(ascending=False)
monthly_play
```

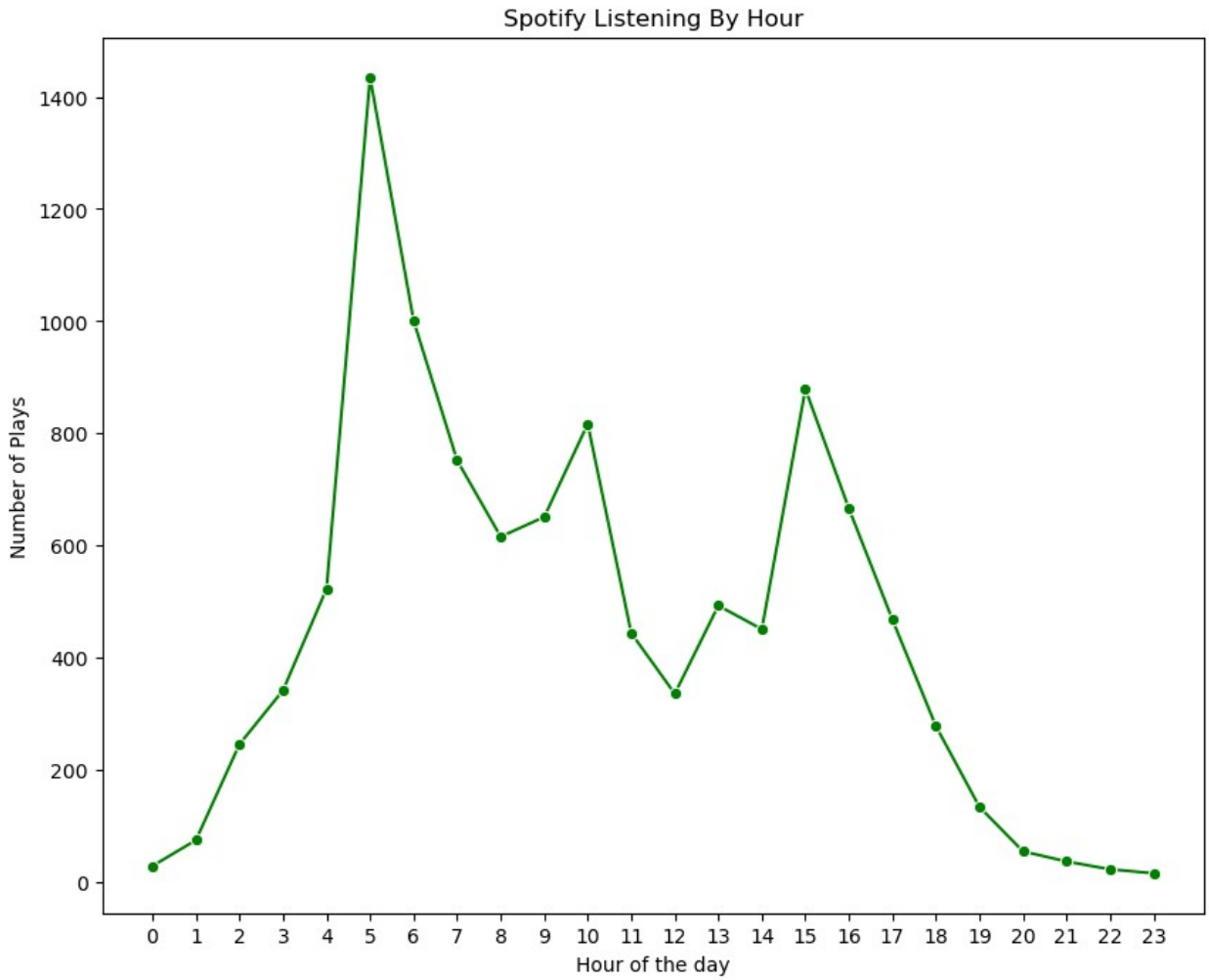
```
Month_name
July         42039.364667
February     28367.830000
June         27257.836833
January      22790.684167
March        22084.580333
May          21823.348833
April        20177.618167
August       19657.471333
September    19243.286833
October      17249.968667
November     13393.817833
Name: ms_played, dtype: float64
```

```
monthly_play= spotify_25.groupby('Month_name')['ms_played'].sum()

plt.figure(figsize=(12,8))
plt.plot(monthly_play, color= 'green', marker= 'o')
plt.title("SpotifyPlays per Month(2025)", color= 'black')
plt.xlabel("Month")
plt.ylabel('Minutes played')
plt.xticks(rotation= 45);
```



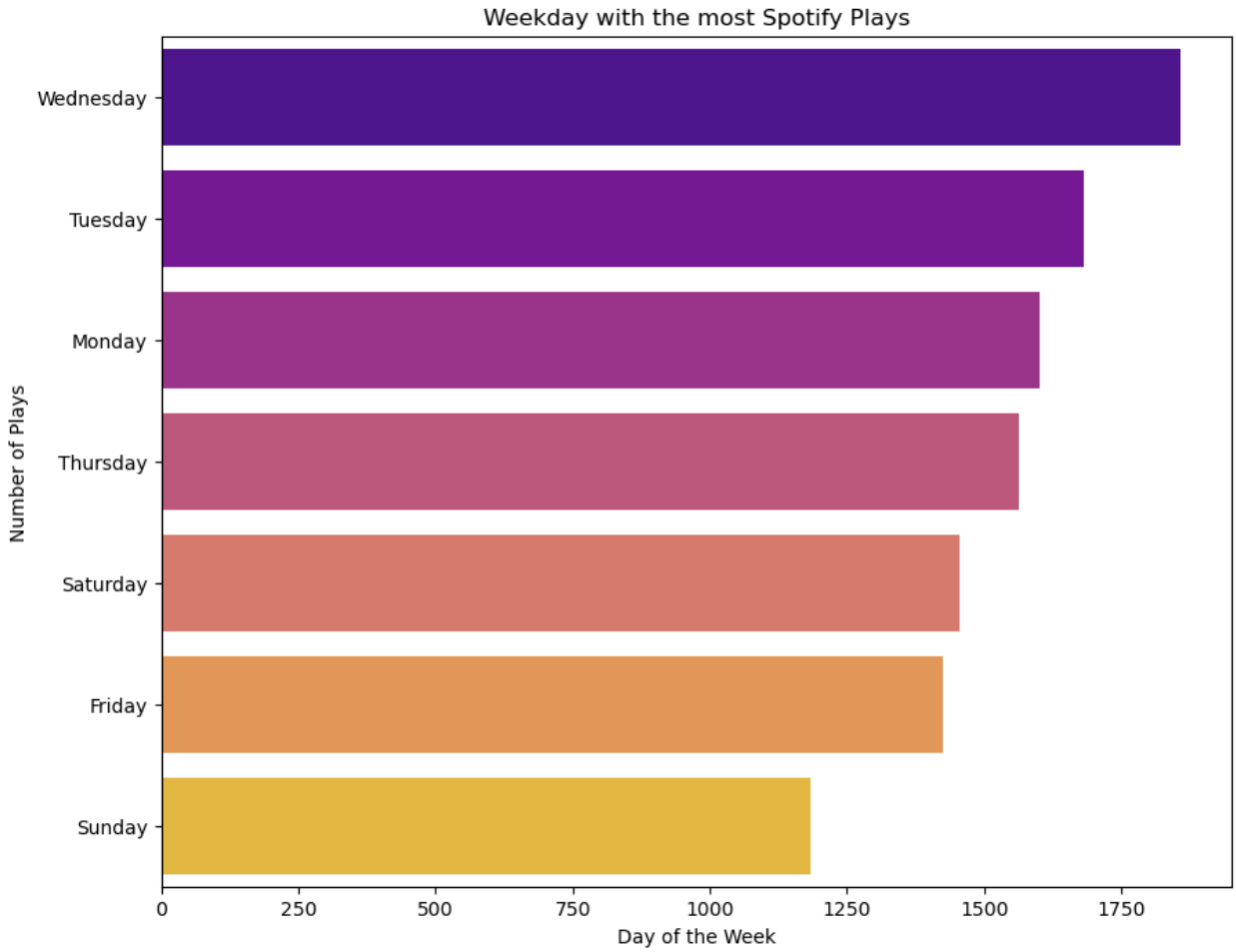
Peak times

```
hourly_play= spotify_25['Hour'].value_counts().sort_index()

plt.figure(figsize=(10,8))
sns.lineplot(y= hourly_play.values, x= hourly_play.index, marker= 'o',
color= 'green')
plt.title("Spotify Listening By Hour")
plt.xlabel('Hour of the day')
plt.ylabel("Number of Plays")
plt.xticks(range(0,24));
```



Day of the week with the most spotify plays

```
spotify_25['Weekday']= spotify_25['Date'].dt.day_name()
plays_weekday=
spotify_25.groupby('Weekday').size().sort_values(ascending=False)
plays_weekday

plt.figure(figsize=(10,8))
sns.barplot(x= plays_weekday.values, y= plays_weekday.index,
palette="plasma")
plt.title("Weekday with the most Spotify Plays")
plt.xlabel("Day of the Week")
plt.ylabel("Number of Plays");
```

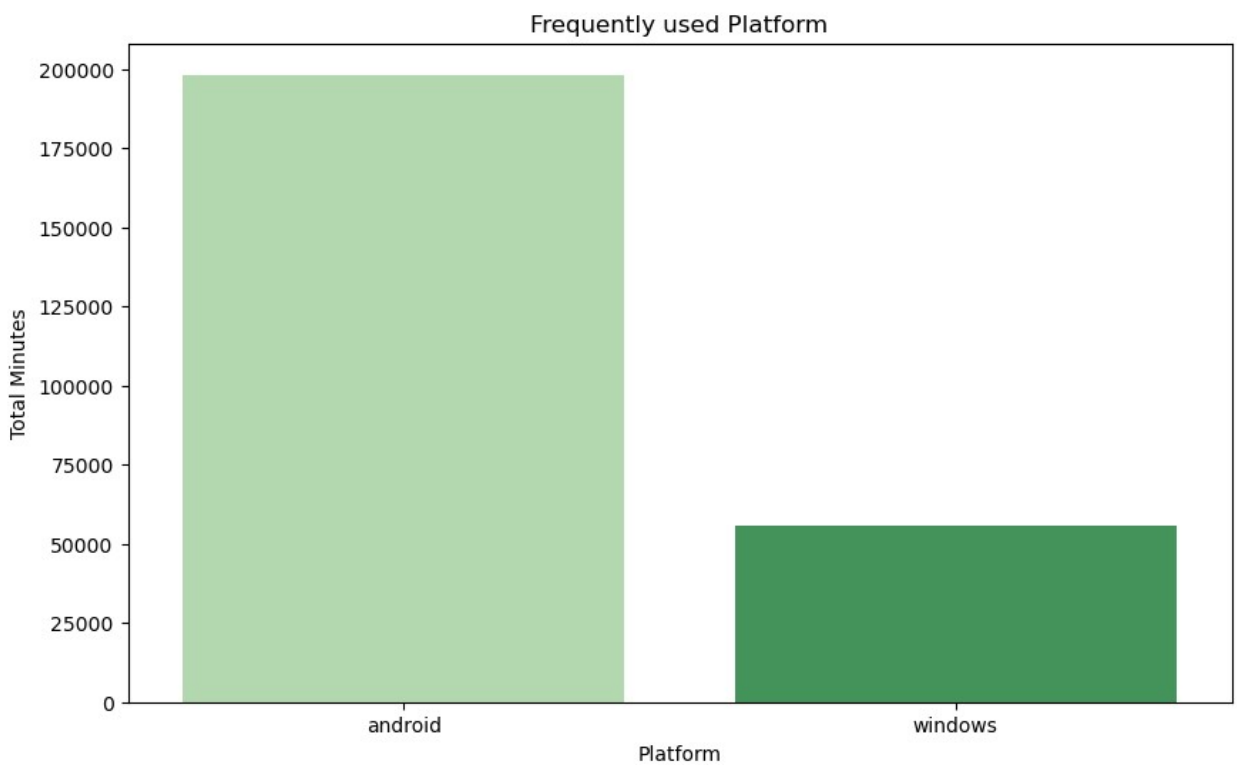



```
spotify_25

                           Date platform  ms_played Country  \
7412  2025-01-01 05:53:50+00:00  android  39.583167      IN
7413  2025-01-01 05:57:25+00:00  android  35.673500      IN
7414  2025-01-01 06:15:31+00:00  android  18.903333      IN
7415  2025-01-01 10:23:32+00:00  android  56.891500      IN
7416  2025-01-01 10:31:03+00:00  android  75.069167      IN
...                         ...      ...        ...     ...
18174 2025-11-18 15:24:36+00:00  android  18.929167      ZW
18175 2025-11-18 15:26:55+00:00  android  22.914000      ZW
18176 2025-11-18 15:26:57+00:00  android   0.349333      ZW
18177 2025-11-18 15:31:04+00:00  android  41.357333      ZW
18178 2025-11-18 15:57:07+00:00  android  32.575833      ZW

                                                Track_name
Album_artist_name  \
7412                                             Esandleni  Nontokozo
Mkhize
7413                                            Kuyabongwa
ChubbyCheeks
```

```
7414                                          Ungukuphila    Xolly
Mncwango
7415             Shayi'Moto (feat. Seemah & Yanda Woods)   Mellow &
Sleazy
7416   Biri Marung (Edit) (feat. Sje Konka, Focalisti...         Mr
Pilato
...                                                     ...
...
18174                              Overloading (OVERDOSE)
Mavins
18175                                             Kwalula   Sun-El
Musician
18176                                        Dangerous II
Usimamane
18177                                    BROWN SKIN GIRL          Blue
Ivy
18178                                         Ngimtholile     Maverick
Muji

                                         Album_album_name  skipped
offline  \
7412                                            Esandleni     True
False
7413                                           Kuyabongwa    False
False
7414                                     Jesus Is Enough    False
False
7415             Shayi'Moto (feat. Seemah & Yanda Woods)     True
False
7416   Biri Marung (Edit) (feat. Sje Konka, Focalisti...     True
False
...                                                     ...      ...
...
18174                                           Chapter X    False
False
18175                                To the World & Beyond     True
True
18176                                        20th: Deluxe     True
True
18177           The Lion King: The Gift [Deluxe Edition]    False
True
18178                                         Ngimtholile    False
True

       Year  Month Month_name  Day  Hour
7412   2025      1    January    1     5
7413   2025      1    January    1     5
7414   2025      1    January    1     6
7415   2025      1    January    1    10
```

```
7416   2025      1    January    1    10
...     ...    ...        ...   ...   ...
18174  2025     11   November   18    15
18175  2025     11   November   18    15
18176  2025     11   November   18    15
18177  2025     11   November   18    15
18178  2025     11   November   18    15

[10767 rows x 14 columns]
```

```
platform_most_used =spotify_25.groupby('platform')
['ms_played'].sum().sort_values(ascending= False)

plt.figure(figsize=(10,6))
sns.barplot(x=platform_most_used.index, y= platform_most_used.values,
palette='Greens')
plt.title("Frequently used Platform")
plt.xlabel("Platform")
plt.ylabel("Total Minutes");
```



Repeat vs new song ratio

```
# Total plays
total_plays = len(spotify_25)

# Unique songs
```

```
unique_songs = spotify_25['Track_name'].nunique()

# Repeat plays
repeat_plays = total_plays - unique_songs

print(f"Repeat plays: {repeat_plays}")
print(f"New songs played: {unique_songs}")
print(f"Repeat vs New ratio: {repeat_plays / unique_songs:.2f}")
```

```
Repeat plays: 8789
New songs played: 1978
Repeat vs New ratio: 4.44
```

Top Artist for Each Month

```
monthly_artist = spotify_25.groupby(['Month', 'Album_artist_name'])
['ms_played'].sum().reset_index()

# Get top artist per month
top_artist_per_month = monthly_artist.loc[
    monthly_artist.groupby('Month')['ms_played'].idxmax()
]

# Sort by month
top_artist_per_month = top_artist_per_month.sort_values('Month')

print(top_artist_per_month)
```

```
      Month   Album_artist_name    ms_played
182       1        Taylor Swift  3588.524500
425       2               Sjava  2468.218000
559       3  Joyous Celebration  2174.869500
831       4               Sjava  1434.851333
992       5               Sjava  2557.238333
1260      6               Sjava  3795.780333
1498      7               Sjava  5078.409667
1734      8               Sjava  3739.165333
1984      9               Sjava  5836.804667
2230     10               Sjava  3590.382167
2431     11               Sjava  5080.241500
```

```
top_5_monthly = (
    spotify_25.groupby(['Month', 'Album_artist_name'])['ms_played']
    .sum()
    .reset_index()
    .sort_values(['Month', 'ms_played'], ascending=[True, False])
    .groupby('Month')
    .head(5)
)
```

```python
print(top_5_monthly)
```

```
      Month   Album_artist_name    ms_played
182       1        Taylor Swift  3588.524500
172       1               Sjava  1807.581167
137       1    Nontokozo Mkhize  1500.596000
174       1             Sneziey   892.383667
205       1      Xolly Mncwango   852.934333
425       2               Sjava  2468.218000
317       2  Joyous Celebration  1454.059167
440       2        Taylor Swift  1150.655667
343       2      Lwah Ndlunkulu  1095.708500
327       2        Khathapillar  1047.824000
559       3  Joyous Celebration  2174.869500
633       3               Sjava  1468.494667
639       3    Spirit Of Praise   712.876833
533       3          Ed Sheeran   710.881500
570       3        Khathapillar   644.495833
831       4               Sjava  1434.851333
801       4    Nontokozo Mkhize  1063.713167
837       4    Spirit Of Praise  1060.153000
750       4  Joyous Celebration   891.479833
840       4        Taylor Swift   822.721000
992       5               Sjava  2557.238333
924       5  Joyous Celebration  2150.587000
995       5    Spirit Of Praise   889.537333
937       5      Lwah Ndlunkulu   811.997000
906       5          Ed Sheeran   782.113167
1260      6               Sjava  3795.780333
1132      6  Joyous Celebration  1688.069667
1052      6        Blaq Diamond  1368.779500
1207      6    Nontokozo Mkhize   735.857667
1063      6           Burna Boy   720.279000
1498      7               Sjava  5078.409667
1398      7  Joyous Celebration  3127.770000
1401      7      Kabza De Small  2956.454667
1499      7             Sneziey  1569.171167
1334      7        Blaq Diamond  1388.163167
1734      8               Sjava  3739.165333
1630      8  Joyous Celebration   813.739167
1634      8      Kabza De Small   703.607167
1749      8        Taylor Swift   656.822667
1735      8             Sneziey   616.160833
1984      9               Sjava  5836.804667
1873      9  Joyous Celebration  1165.103000
1990      9    Spirit Of Praise  1014.840833
1994      9        Taylor Swift   448.776667
1946      9    Nontokozo Mkhize   445.835833
2230     10               Sjava  3590.382167
```

```
2132     10  Joyous Celebration  1282.431333
2244     10        Taylor Swift   647.666333
2066     10        Blaq Diamond   600.016333
2197     10    Nontokozo Mkhize   586.001333
2431     11               Sjava  5080.241500
2440     11        Taylor Swift   990.732000
2403     11        Nomfundo Moh   267.995000
2455     11           Usimamane   259.081833
2347     11        Inkabi Zezwe   232.236333
```

```
pivot_artists = top_5_monthly.pivot(
    index='Month',
    columns='Album_artist_name',
    values='ms_played'
).fillna(0)

print(pivot_artists)
```

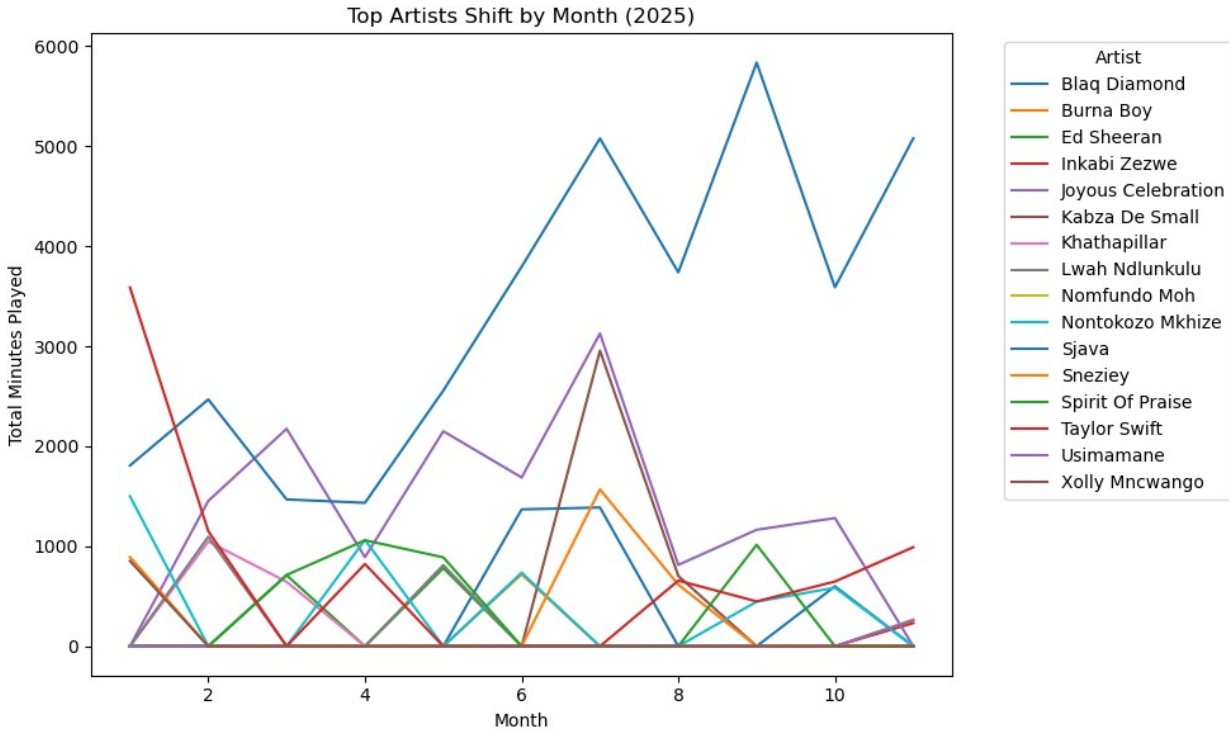
```
Album_artist_name  Blaq Diamond  Burna Boy  Ed Sheeran  Inkabi
Zezwe  \
Month

1                      0.000000      0.000    0.000000      0.000000

2                      0.000000      0.000    0.000000      0.000000

3                      0.000000      0.000  710.881500      0.000000

4                      0.000000      0.000    0.000000      0.000000

5                      0.000000      0.000  782.113167      0.000000

6                   1368.779500    720.279    0.000000      0.000000

7                   1388.163167      0.000    0.000000      0.000000

8                      0.000000      0.000    0.000000      0.000000

9                      0.000000      0.000    0.000000      0.000000

10                   600.016333      0.000    0.000000      0.000000

11                     0.000000      0.000    0.000000    232.236333


Album_artist_name  Joyous Celebration  Kabza De Small  Khathapillar  \
Month
1                           0.000000        0.000000      0.000000
2                        1454.059167        0.000000   1047.824000
3                        2174.869500        0.000000    644.495833
4                         891.479833        0.000000      0.000000
```

```
5                          2150.587000        0.000000      0.000000
6                          1688.069667        0.000000      0.000000
7                          3127.770000     2956.454667      0.000000
8                           813.739167      703.607167      0.000000
9                          1165.103000        0.000000      0.000000
10                         1282.431333        0.000000      0.000000
11                            0.000000        0.000000      0.000000

Album_artist_name  Lwah Ndlunkulu  Nomfundo Moh  Nontokozo Mkhize  \
Month
1                           0.0000         0.000       1500.596000
2                        1095.7085         0.000          0.000000
3                           0.0000         0.000          0.000000
4                           0.0000         0.000       1063.713167
5                         811.9970         0.000          0.000000
6                           0.0000         0.000        735.857667
7                           0.0000         0.000          0.000000
8                           0.0000         0.000          0.000000
9                           0.0000         0.000        445.835833
10                          0.0000         0.000        586.001333
11                          0.0000       267.995          0.000000

Album_artist_name        Sjava      Sneziey  Spirit Of Praise  Taylor
Swift  \
Month

1                   1807.581167   892.383667          0.000000
3588.524500
2                   2468.218000     0.000000          0.000000
1150.655667
3                   1468.494667     0.000000        712.876833
0.000000
4                   1434.851333     0.000000       1060.153000
822.721000
5                   2557.238333     0.000000        889.537333
0.000000
6                   3795.780333     0.000000          0.000000
0.000000
7                   5078.409667  1569.171167          0.000000
0.000000
8                   3739.165333   616.160833          0.000000
656.822667
9                   5836.804667     0.000000       1014.840833
448.776667
10                  3590.382167     0.000000          0.000000
647.666333
11                  5080.241500     0.000000          0.000000
990.732000

Album_artist_name   Usimamane  Xolly Mncwango
```

```
Month
1                     0.000000      852.934333
2                     0.000000        0.000000
3                     0.000000        0.000000
4                     0.000000        0.000000
5                     0.000000        0.000000
6                     0.000000        0.000000
7                     0.000000        0.000000
8                     0.000000        0.000000
9                     0.000000        0.000000
10                    0.000000        0.000000
11                  259.081833        0.000000
```

Top Artists Shift by Month (2025)

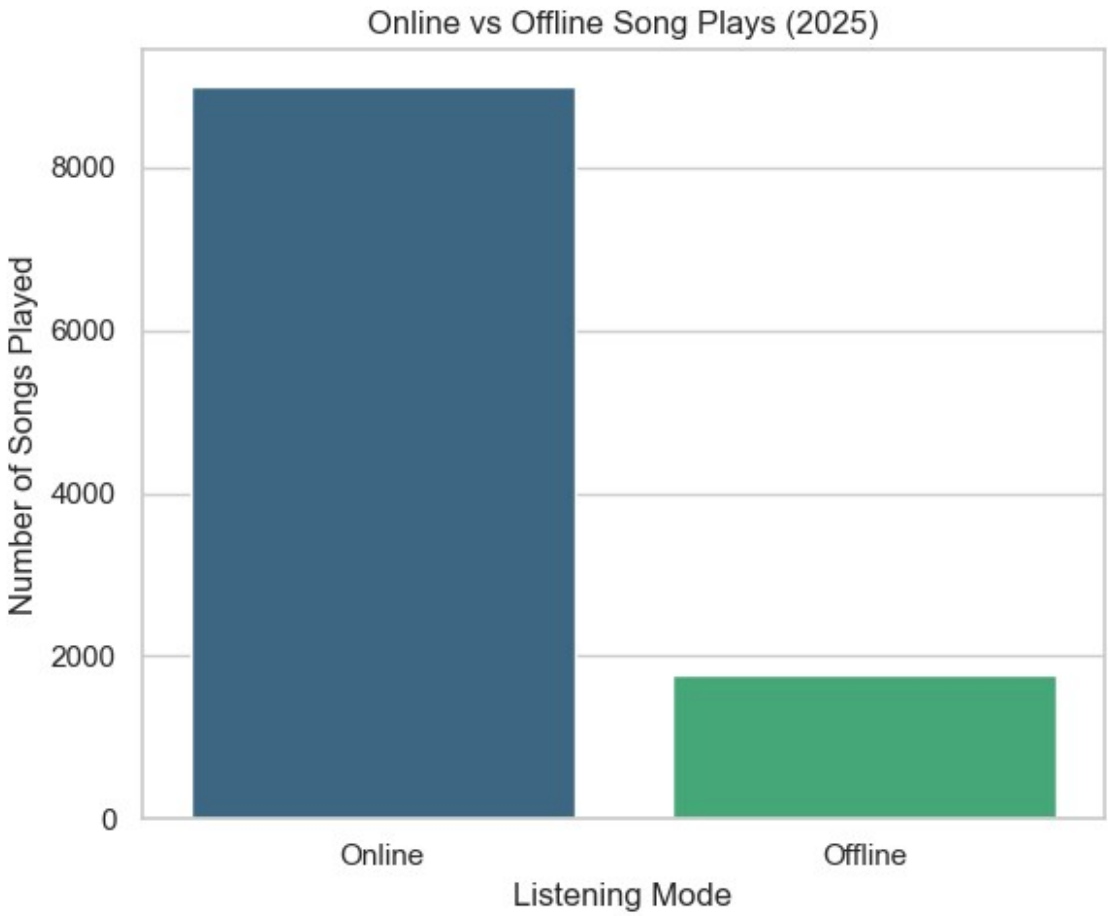
```
import matplotlib.pyplot as plt

pivot_artists.plot(figsize=(10,6))
plt.title('Top Artists Shift by Month (2025)')
plt.xlabel('Month')
plt.ylabel('Total Minutes Played')
plt.legend(title='Artist', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



```python
#Number of Songs Played online vs Offline
online_offline_counts =
spotify_25['offline'].value_counts().reset_index()
online_offline_counts.columns = ['Offline Status', 'Number of Songs']

# Replace True/False with labels
online_offline_counts['Offline Status'] =
online_offline_counts['Offline Status'].replace({
    True: 'Offline',
    False: 'Online'
})

print(online_offline_counts)
```

```
  Offline Status  Number of Songs
0         Online             8997
1        Offline             1770
```

```python
sns.set(style="whitegrid")

plt.figure(figsize=(6,5))
sns.barplot(
    data=online_offline_counts,
    x='Offline Status',
    y='Number of Songs',
    palette='viridis'
)

plt.title("Online vs Offline Song Plays (2025)")
plt.xlabel("Listening Mode")
plt.ylabel("Number of Songs Played")
plt.tight_layout()
plt.show()
```




```
Country_listen = spotify_25['Country'].value_counts().reset_index()
Country_listen
```

```
  Country  count
0      ZW   6239
1      IN   3264
2      IT    545
3      MU    301
4      DE    146
5      FR    129
6      ES     93
7      US     44
8      CH      6
```

```
#Saving my new dataset
spotify_25.to_csv(r"c:\Users\My PC\Desktop\Spotify Wrapped 2025\
spotify_2025_analysis.csv", index=False)

print("Dataset successfully saved !!!!!!!!!!!!!")
```

```
Dataset successfully saved !!!!!!!!!!!!!
```